DR. A.P.J. ABDUL KALAM TECHNICAL UNIVERSITY, LUCKNOW

Evaluation Scheme & Syllabus
For
B.Tech. 2nd Year

Plastic Engineering

(Effective from the Session: 2023-24)

DR. A.P.J. ABDUL KALAM TECHNICAL UNIVERSITY, LUCKNOW

SEMESTER –III

SN	Subject Code	Subject	Type	Category	Periods			Sessional Component		Sessional (SW) (TS/PS)	End Semester Examination (ESE)	Total SW+ESE	Credit Cr
					L	T	P	CT	TA	CT+TA	TE/PE		
1	BOE3** / BAS303	Science Based Open Elective/BSC (Maths-III/Math IV/ Math V)	T	ES/BS	3	1	0	20	10	30	70	100	4
2	BVE301 / BAS301	Universal Human Value and Professional Ethics/ Technical Communication	T	VA/HS	2	1	0	20	10	30	70	100	3
3	BME301	Thermodynamics	T	PC	3	1	0	20	10	30	70	100	4
4	BPL301	Fluid Mechanics & Rheology	T	PC	3	1	0	20	10	30	70	100	4
5	BME303	Materials Engineering	T	PC	2	1	0	20	10	30	70	100	3
6	BME351	Fluid Mechanics Lab	P	PC	0	0	2		50	50	50	100	1
7	BME352	Material Testing Lab	P	PC	0	0	2		50	50	50	100	1
8	BME353	Computer Aided Machine Drawing-I Lab	P	PC	0	0	2		50	50	50	100	1
10	BCC301 / BCC302	Cyber Security/Python programming	T	VA	2	0	0	20	10	30	70	100	2
11	BCC351	Internship Assessment /Mini Project*	P							100		100	2
		Total			**15**	**5**	**6**						**25**

- **Mathematics –III** for CE / ENV and allied branches
- **Mathematics-IV** for Computer/Electronics/Electrical & allied Branches, Mechanical & Allied Branches Textile/Chemical & allied Branches
- **Mathematics-V** for Bio Technology / Agriculture Engineering

SEMESTER –IV

SN	Subject Code	Subject	Type	Category	Periods			Sessional Component		Sessional (SW) (TS/PS)	End Semester Examination (ESE)	Total SW+ESE	Credit Cr
					L	T	P	CT	TA	CT+TA	TE/PE		
1	BAS403 / BOE4**	BSC (Maths-III/Math IV/ Math V)/Science Based Open Elective	T	BS/ES	3	1	0	20	10	30	70	100	4
2	BAS401 / BVE401	Technical Communication / Universal Human Value and Professional Ethics	T	HS/VA	2	1	0	20	10	30	70	100	3
3	BPL401	Plastic Materials-I	T	PC	3	1	0	20	10	30	70	100	4
4	BPL402	Plastic Processing Technology	T	PC	3	1	0	20	10	30	70	100	4
5	BPL403	Fundamental of Polymer Science	T	PC	2	1	0	20	10	30	70	100	3
6	BPL451	Synthesis & Polymerization Lab	P	PC	0	0	2		50	50	50	100	1
7	BPL452	Plastics Processing Lab	P	PC	0	0	2		50	50	50	100	1
8	BPL453	Identification & Polymers Lab	P	PC	0	0	2		50	50	50	100	1
9	BCC402 / BCC401	Python Programming/Cyber Security	P	VA	2	0	0	20	10	30	70	100	2
10	BVE451 / BVE452	Sports and Yoga - II / NSS-II	P	VA	0	0	3			100		100	0
		Total			**15**	**5**	**9**						**23**
		Minor Degree/ Honors Degree MT-1/HT-1											

*****The Mini Project or internship (4 weeks) will be done during summer break after 4th Semester and will be assessed during V semester.

SEMESTER-III

BME301- THERMODYNAMICS

Objectives: **L-T-P : 3-1-0**

CO1: To understand the basic terms of thermodynamics: **K2**
CO2: To apply I law to various energy conversion devices: **K3**
CO3: To evaluate the changes in properties of substances in various processes: **K3**
CO4: To understand the difference between high grade and low-grade energies: **K2**

UNIT I (08 Lecture)

Review of Fundamental Concepts and Definitions:

Introduction- Basic Concepts: System, Control Volume, Surrounding, Boundaries, Universe, Types of Systems, Macroscopic and Microscopic viewpoints, Concept of Continuum, Thermodynamic Equilibrium, State, Property, Process, Exact & Inexact Differentials, Cycle Reversibility Quasi – static Process, Irreversible Process, Causes of Irreversibility Energy and its forms, Work and heat (sign convention), Gas laws, Ideal gas, Real gas, Law of corresponding states, Property of mixture of gases, electrical, magnetic, gravitational, spring and shaft work.

Zeroth law of thermodynamics: Concept of Temperature and its' measurement, Different Temperature scales.

UNIT II (08 Lecture)

First law of thermodynamics:

First Law for closed system, Work and Heat Transfer in various processes, Internal Energy, Enthalpy, Derivation of General energy equation for a control volume; Steady Flow Energy Equation, Examples of steady flow devices: Boilers, Nozzles, Condensers, Turbine, Throttling Device, Pumps etc.; Unsteady processes and its examples, Limitations of first law of thermodynamics, PMM-I.

UNIT III (08 Lecture)

Second law of thermodynamics:

Thermal reservoirs, Energy conversion, Heat engines, Efficiency, Reversed heat engine, Heat pump, Refrigerator, Coefficient of Performance, Kelvin Planck and Clausius statement of second law of thermodynamics, Equivalence of the two statements. Reversible and irreversible processes, Carnot cycle and Carnot engine, Carnot theorem and it's corollaries, Thermodynamic Temperature Scale, PMM-II. **Entropy:** Clausius inequality, Concept of Entropy, Entropy change of pure substance in different thermodynamic processes, TdS equation, Principle of entropy increase, T-S diagram, Statement of the third law of thermodynamics.

UNIT IV (08 Lecture)

Availability and Irreversibility:

Available and unavailable energy, Availability and Irreversibility, Second law efficiency

Thermodynamic Relations:

Conditions for exact differential, Helmholtz & Gibb's function, Maxwell relations, Clapeyron equation, Joule-Thompson coefficient and Inversion curve. Coefficient of volume expansion, Adiabatic and Isothermal compressibility.

UNIT V (08 Lecture)

Properties of steam and Rankine cycle:

Pure substance, Property of Pure Substance (steam), Triple point, Critical point, Saturation states, Sub- cooled liquid state, Superheated vapour state, Phase transformation process of water, Graphical representation of pressure, volume and temperature, P-T, P-V and P-h diagrams, T-S and H-S diagrams, use of property diagram, Steam-Tables &Moller chart, Dryness factor and it's measurement, processes involving steam in closed and open systems. Simple Rankine cycle.

Course Outcomes:

1. After completing this course, the students will be able to apply energy balance to systems and control volumes, in situations involving heat and work interactions.
2. Students can evaluate changes in thermodynamic properties of substances.
3. The students will be able to evaluate the performance of energy conversion devices.
4. The students will be able to differentiate between high grade and low-grade energies.

Books and References:

1. Basic and Applied Thermodynamics by P.K Nag, MCGRAW HILL INDIA.
2. Thermodynamics for Engineers by Kroos & Potter, Cengage Learning.
3. Thermodynamics by Shavit and Gutfinger, CRC Press.
4. Thermodynamics- An Engineering Approach by Cengel, MCGRAW HILL INDIA.
5. Basic Engineering Thermodynamics, Joel, Pearson.
6. Fundamentals of Engineering Thermodynamics by Rathakrishnan, PHI.
7. Engineering Thermodynamics by Dhar, Elsevier.
8. Engineering Thermodynamics by Onkar Singh, New Age International.
9. Engineering Thermodynamics by CP Arora.
10. Engineering Thermodynamics by Rogers, Pearson.
11. Thermodynamics by Prasanna Kumar, Pearson Education
12. Engineering Thermodynamics by Mishra, Cengage Learning.

BPL302- FLUID MECHANICS AND RHEOLOGY		
Course Outcome: Student will be able to		**Blooms Taxonomy**
CO 1	Learn about the application of mass and momentum conservation laws for fluid flows.	**K2**
CO 2	Understand the importance of dimensional analysis.	**K2**
CO 3	Learn about velocity and pressure variations in various types of simple flows.	**K2**
CO 4	Learn the basics of polymer deformation and flow properties.	**K2**
CO 5	Analyses the rheological properties of polymer systems using rheometers .	**K3**

UNIT-I (08 Lectures)
Definition of fluid, Newton's law of viscosity, Units and dimensions-Properties of fluids, mass density, specific volume, specific gravity, viscosity, compressibility and surface tension, Incompressible flow, Bernoulli's equation and its applications - Pitot tube, orifice meter, venturi meter and bend meter, notches and weirs, momentum equation and its application to pipe bends.

UNIT-II (08 Lectures)
Continuum & free molecular flows. Steady and unsteady, uniform and non-uniform, laminar and turbulent flows, rotational and irrotational flows, compressible and incompressible flows, subsonic, sonic and supersonic flows, sub-critical, critical and supercritical flows, one, two- and three dimensional flows, streamlines, continuity equation for 3D and 1D flows, circulation, stream function and velocity potential. Buckingham's Pi theorem, important dimensionless numbers and their significance.

UNIT-III (08 Lectures)
Equation of motion for laminar flow through pipes, turbulent flow, isotropic, homogenous turbulence, scale and intensity of turbulence, measurement of turbulence, eddy viscosity, resistance to flow, minor losses, pipe in series and parallel, power transmission through a pipe, siphon, water hammer, three reservoir problems and pipe networks.
Boundary layer thickness, boundary layer over a flat plate, laminar boundary layer, application of momentum equation, turbulent boundary layer, laminar sublayer, separation and its control, Drag and lift, drag on a sphere, a two-dimensional cylinder, and an aerofoil, Magnus effect.

UNIT-IV (08 Lectures)
Introduction to polymer rheology, Newtonian and non-Newtonian fluids, shear stress, shear strain and shear rate, shear modulus, bulk modulus, Zero shear viscosity, Dependence of viscosity with temperature, shear stress, Viscoelasticity - effect of rate of strain, temperature and time on mechanical behavior of polymeric materials.

UNIT-V(08 Lectures)
Measurement of viscosity and normal stresses. Dynamic flow behavior, time dependent fluid responses. Capillary rheometers, cone and plate viscometer, parallel-plate rheometer, oscillating disc rheometer, Mooney viscometer. Rheology of modified polymer systems. Measurements of rheological properties.

Books and References:

1. Introduction to fluid mechanics and Fluid machines by S.K Som, Gautam Biswas, S Chakraborty.
2. Fluid mechanics and machines by R.K Bansal.
3. F. M. White, Fluid Mechanics, 6th Ed., Tata McGraw-Hill, 2008.
4. Fluid Mechanics and Its Applications by V.K. Gupta et.al. 5. Fluid Mechanics by Yunus Cengel
5. Batchelor, G. K. (1999). Introduction to fluid dynamics. New Delhi, India: Cambridge University Press.
6. Acheson, D. J. (1990). Elementary fluid dynamics. New York, USA: Oxford University Press
7. R.W. Fox, A.T. McDonald and P.J. Pritchard, Introduction to Fluid Mechanics, 6th Ed., John Wiley, 2004.
8. Fluid Mechanics and Hydraulic Machines, 5/ed by Mahesh Kumar, Pearson Education
9. Fluid Mechanics, 2e in SI Units, by R.C Hibbeler, Pearson Education

BME303- MATERIALS ENGINEERING

CO-1 : Students will be able to identify the crystal structure and measure the mechanical properties of materials. K3
CO-2 : Students will be able to test the various failures of materials. K3
CO-3 : Students will be able to identify the mechanical properties based on composition of micro-constituents depicted in the phase-diagram. K3
CO-4 : Students will understand the concept of improving the mechanical properties through heat treatment. K2
CO-5 : Students will learn the structure and properties of alloys and composites. K2

UNIT-I (08 Lecture)

Crystal Structure: Unit cells, Metallic crystal structures, Ceramics. Imperfection in solids: Point, line, interfacial and volume defects; dislocation strengthening mechanisms and slip systems, critically resolved shear stress. Mechanical Properties and its measurement: Tensile, compression and torsion tests; Young's modulus, relations between true and engineering stress-strain curves, generalized Hooke's law, yielding and yield strength, ductility, resilience, toughness and elastic recovery; Hardness: Rockwell, Brinell and Vickers and their relation to strength.

UNIT-II (08 Lecture)

Alloys, substitutional and interstitial solid solutions- Hume-Rothery Rule, Phase diagrams: Interpretation of binary phase diagrams and micro-structure development; Lever Rule, eutectic, eutectoid, peritectic, peritectoid and monotectic reactions. Iron and Iron-carbide phase diagram, microstructural aspects of ledeburite, austenite, ferrite and cementite, cast iron, High Speed Steel

UNIT-III (08 Lecture)

Alloying of steel, properties of stainless steel and tool steels, maraging steels- cast irons; grey, white, malleable and spheroidal cast irons- copper and copper alloys; brass, bronze and cupro-nickel; Aluminium and Al-Cu – Mg alloys- Nickel based superalloys and Titanium alloys, Carbon Fibre and alloys, Structure and properties of Graphene Material, Composite Materials, Reinforcement

UNIT-IV (08 Lecture)

Heat treatment of Steel: Annealing, tempering, normalising and spheroidising, isothermal transformation diagrams for Fe-C alloys and microstructure development. Continuous cooling curves and interpretation of final microstructures and properties austempering, martempering, case hardening, carburizing, nitriding, cyaniding, carbo-nitriding, flame and induction hardening, vacuum and plasma hardening.

UNIT-V (08 Lecture)

Creep and failures, Introduction to non-destructive testing (NDT): Visual Inspection, Liquid Penetrate Testing, Magnetic Particle Inspection, Ultra-sonic Testing, Acoustic Testing, Radiography, X-ray Differaction, Eddy Current Testing etc., Material behaviour / properties at a very low (Cryogenics) and high temperature.

Books and References:

1. W. D. Callister, 2006, "Materials Science and Engineering-An Introduction", 6th Edition, Wiley
2. Kenneth G. Budinski and Michael K. Budinski, "Engineering Materials", Prentice Hall of India Private Limited, 4th Indian Reprint, 2002.
3. V. Raghavan, "Material Science and Engineering', Prentice Hall of India Private Limited, 1999.
4. Mechanics of materials by James M. Gere.
5. Introduction to engineering materials by B.K. Agarwal.
6. Physical metallurgy and advanced materials by R.E. Smallman.
7. Engineering mechanics of composite materials by Isaac M. Daniel.
8. U. C. Jindal, "Engineering Materials and Metallurgy", Pearson, 2011.
9. Introduction to Materials Science For Engineers by Shackelford, 9th edn, 2023, Pearson Education
10. Engineering Materials: Properties and Selection, 9edn by Budinski, Pearson Education

BME351- FLUID MECHANICS LAB

Course Outcome (CO)		Bloom's Knowledge Level (KL)
At the end of course , the student will be able to		
CO 1	Understand the principles and performance characteristics of flow and thermal devices.	**K2**
CO 2	Know about the measurement of the fluid properties	**K1**
CO 3	Understand and analyze various properties of fluids	**K3**
CO 4	Evaluate the performance characteristics of fluid/thermal machinery	**K3**
CO 5	Evaluate the velocity and pressure variations in various types of simple flows.	**K3**
DETAILED SYLLABUS		**0-0-2**

List of Experiments:(At least 8 of the following)

1. To determine the coefficient of impact for vanes.
2. To determine coefficient of discharge of an orifice meter.
3. To determine the coefficient of discharge of Notch (V and Rectangular types).
4. To determine the friction factor for the pipes.
5. To determine the coefficient of discharge of venturi meter.
6. To determine the coefficient of discharge, contraction & velocity of an orifice.
7. To verify the Bernoulli's Theorem.
8. To find critical Reynolds number for a pipe flow.
9. To determine the meta-centric height of a floating body.
10. To determine the minor losses due to sudden enlargement, sudden contraction and bends.
11. To show the velocity and pressure variation with radius in a forced vortex flow.

BME352- MATERIALS TESTING LAB

CO-1 : Students will be able to perform different destructive and non-destructive testing methods to measure various mechanical properties. **K2**
CO-2 : Students will be able to analyse the effect of different heat-treatment processes on the Hardness. **K4**
CO-3 : Students will be able to simulate the material using simulating software / measure the mechanical properties of 3-D printed components. **K3**

List of Experiments: (At least 8 of the following)

1. Strength test of a given mild steel specimen on UTM with full details and stress versus strain plot on the machine.
2. Other tests such as shear, bend tests on UTM.
3. Impact test on impact testing machine like Charpy, Izod or both.
4. Hardness test of given specimen using Rockwell and Vickers/Brinell testing machines (this test should be conducted on MS Specimen – prior and after the various heat-treatment processes like Annealing / Normalizing / Hardening etc.; with this, 2 experiments can be designed: one for Annealing or Normalizing and another for Hardening / Tempering).
5. Fatigue test on fatigue testing machine.
6. Creep test on creep testing machine.
7. Study of NDT (non-destructive testing) methods like magnetic flaw detector, ultrasonic flaw detector, eddy current testing machine, dye penetrant tests.
8. Modelling of Graphene / any other materials structure using LAMMPs (LAMMPS Molecular Dynamics Simulator) or any other simulation software and studying the mechanical properties.
9. Measurement of mechanical properties (tensile test, compression test, hardness test / flexural test) of specimen made by 3-D printing

BME353- COMPUTER AIDED MACHINE DRAWING-I LAB

Course Outcome (CO)		Bloom's Knowledge Level (KL)
At the end of course , the student will be able to		
CO 1	Understand and apply 2D software to develop a part model	**K3**
CO 2	Understand about temporary and permanent fasteners	**K2**
CO 3	Understand the need for free hand sketching, Free hand sketching of foundation bolts etc.	**K2**
CO 4	Create assembly drawing of simple machine elements like rigid or flexible coupling	**K3**
CO 5	Create 2D drawings and assemblies of various machine components	**K3**
DETAILED SYLLABUS		**0-0-2**

UNIT-I

Computer aided drafting (1 drawing) Introduction to computer aided drafting; advantages and applications of CAD, concepts of computer aided 2D drafting using any drafting software like AutoCAD, Pro-E, Solid Edge, Draft Sight etc., basic draw and modify commands, making 2D drawings of simple machine parts. Course Outcomes: Upon completion of this course, the students can use computer and CAD software for modelling mechanical components.

UNIT-II

Fasteners (2 drawing sheets) Temporary and permanent fasteners, thread nomenclature and forms, thread series, designation, representation of threads, bolted joints, locking arrangement of nuts, screws, washers, foundation bolts etc., keys, types of keys, cotter and knuckle joints.

UNIT-III

Riveted joints (1 drawing sheet) Introduction, rivets and riveting, types of rivets, types of riveted joints, drawing of boiler joints etc. Free hand sketching (1 drawing sheet) Introduction, need for free hand sketching, Free hand sketching of foundation bolts, studs, pulleys, couplings etc.

UNIT-IV

Assembly drawing (2 drawing sheets) Introduction to assembly drawing, drawing assembly drawing of simple machine elements like rigid or flexible coupling, muff coupling, Plummer block, footstep bearing, bracket etc.

Books and References:

1. Fundamentals of Machine Drawing by Sadhu Singh & Shah, PHI.
2. Engineering Drawing by Bhat, & Panchal, Charotar Publishing House.
3. Machine Drawing with AutoCAD by Pohit and Ghosh, Pearson.
4. Machine Drawing-KL Narayana, P Kannaiah, KV Reddy, New Age.
5. Machine Drawing, N. Siddeshswar, P Kannaiah, VVS Shastry, Tata McGraw Hill.
6. Engineering Drawing, Pathak, Wiley.
7. Textbook of Machine Drawing, K C John, PHI.
8. AutoCAD 2014 for Engineers & Designers, Bhatt, WILEY
9. Engineering Drawing, 2nd edn by Shah & Rana
10. Fundamentals of Engineering Drawing, 11th edn by Warren J. Luzadder, Pearson Education

BPL401- PLASTICS MATERIALS –I		
Course Outcome: Student will be able to		**Blooms Taxonomy**
CO 1	Understand the basics of Plastics and their classification	**K3**
CO 2	Understand various concepts of structure of Plastics	**K3**
CO 3	Understand the basics of Polyolefins and their copolymers	**K3**
CO 4	Understand the basics of Styrene based Polymers	**K2**
CO 5	Understand the basics of Epoxy Polymers	**K2**
CO 6	Understand the basics of Biodegradable Polymers and their applications	**K3**

S.No.	CONTENTS
1	**UNIT I (08 Lecture)** Plastics: Definition and Classification of Plastics, General properties and types, Thermoplastics, Thermosetting, Engineering & High-performance plastics. Structure of Plastics: Molecules, Chains, Crystallinity and its effect on properties, Crosslinked plastics, Determination of Molecular weight, Effect of Molecular weight on processing and properties, Molecular weight distribution. Linear, Branched and cross-linked structures in polymers. Glass transition temperature (Tg).
2	**UNIT II (08 Lecture)** Polyolefins and their copolymers: Introduction, Sources and manufacture of raw materials, basic chemistry, Methods of manufacture, Flow behavior, General properties and applications. Vinyl chloride polymers and co-polymers: Introduction, Sources and manufacture of raw materials, basic chemistry, Methods of manufacture, Flow behavior, General properties and applications.
3	**UNIT III (08 Lecture)** Styrene and Styrene co- polymers: Introduction, Sources and manufacture of raw materials, basic chemistry, Methods of manufacture, Flow behavior, General properties and applications. Acrylate Polymers: Introduction, Sources and manufacture of raw materials, basic chemistry, Methods of manufacture, Flow behavior, General properties and applications. Cellulose polymers: History, Development, Sources and manufacture of raw materials, basic chemistry, Methods of manufacture, General properties and applications.
4	**UNIT IV (08 Lecture)** Epoxies: Introduction, Sources and manufacture of raw materials, basic chemistry, Methods of manufacture, Flow behavior, General properties and applications. Polyurethane (PU): Introduction, Sources and manufacture of raw materials, basic chemistry, Methods of manufacture, Processing behaviour, General properties and applications. Silicones: Introduction, Sources and manufacture of raw materials, basic chemistry, Methods of manufacture, Processing behaviour, General properties and applications.
5	**UNIT V (08 Lecture)** Bio degradable and Bio Plastics: Principle and Mechanism of Plastics degradation, Natural Bio-degradable Polymers, Synthetic Biodegradable Polymers, Water soluble Polymers. Bio plastics: Types, properties, and applications.

Books and References:

1. Brydson, J.A “Plastics Material”
2. Schwartz & Good man “Plastics materials and processing”
3. Irwin Rubin “Hand book of Plastic Materials and Technology

BPL402- PLASTICS PROCESSING TECHNOLOGY – I		
Course Outcome: Student will be able to		**Blooms Taxonomy**
CO 1	Understand the basics of Processing of Plastics	**K2**
CO 2	Understand & analyze the process of extrusion	**K3**
CO 3	Understand the application of extrusion in manufacturing of various products	**K3**
CO 4	Understand the concepts Compression & Transfer Moulding and their applications	**K2**
CO 5	Understand the process of Calendaring and its applications	**K2**
CO 6	Analyze the need for auxiliary equipment and their use in various processes	**K3**

S.No.	CONTENTS
1	**UNIT I (08 Lecture)** Introduction to Plastic processing – The need for Plastic Processing, Definition and Fundamentals of Plastic Processing. Classification of Processing methods, Plastics processing techniques, Processability, Effect of Polymer properties on Processing, Process selection.
2	**Unit II (08 Lecture)** General description of extrusion processes, purpose, advantages, disadvantages. Equipment, Normal operation and control, Material and Product considerations, Types of extruder screw, Calculation of Extruder output, influence of screw dimensions on output, design of barrel and screw for commodity, heat sensitive and engineering polymers. Barrier Screws.
3	**Unit III (08 Lecture)** Individual extrusion systems, Dies, Sizing and Downstream equipment, Faults, Causes and Remedies for Film, Pipe, Lamination, Profiles, Cables, Sheet, Box Strapping, Twin-screw extrusion and Co Extrusion systems. Casting of films. Multilayer systems for Films and Pipe. Faults, causes & remedies.
4	**Unit IV (08 Lecture)** General description of Compression and Transfer moulding, Materials, Processing characteristics, Preheating, Mold Heating, Outgassing, Postcuring, Insert Moulding, Various types of Moulds and machines, Shrinkage, Faults, Causes & Remedies. Design Analysis, Cost.
5	**UNIT V (08 Lecture)** Calendaring: Introduction, calendar roll, calendar configuration and operations. Materials, Calendar Films, Fluxing and Feeding, Processing conditions, Design aspects, surface finishing, process controls, advantage of Calendaring over extrusion, Troubleshooting, Auxiliary equipment, Material Handling, Energy conservation, Parts Handling, Drying, Feeders & Blenders, Granulators.

Books and References

1. Plastics Processing Data Handbook by Dominick Rosato, Chapman & Hill
2. Principles of Polymer Processing by Zehev Tadmor &Costas G. Gogos, John Wiley & Sons.
3. Plastics engineering hand book by Berins, Society of the Plastics industry
4. Polymer Extrusion by Chris Rauwendaal, Hanser Publication
5. Handbook of Plastic Processes by Harper, Wiley Interscience
6. Plastics: Materials and Processing, A. Brent Strong , Pearson

BPL403- FUNDAMENTALS OF POLYMER SCIENCE

Course Outcome: Student will be able to		Blooms Taxonomy
CO 1	Understand the basic of Polymers and their properties	K2
CO 2	Understand various processes of Polymerization	K2
CO 3	Understand the concepts of Molecular weight and its significance	K3
CO 4	Understand the properties of Polymers and their significance	K2
CO 5	Be able to understand the degradation of Polymers and apply the knowledge in practical applications	K3

S. No.	CONTENTS
1	**UNIT I (08 Lecture)** History of polymer science, Classification of Polymers, Functionality and structure of polymers, Physical properties and characterization of polymers, effect of structure on properties of polymers, Inorganic polymers. Concept of macromolecules. Stereochemistry of polymers.
2	**UNIT II (08 Lecture)** Introduction, Chain & Step growth polymerization, Polymerization techniques, Kinetics of Polymerization (Free radical, Cationic, Anionic polymerization, Polycondensation).
3	**UNIT III (08 Lecture)** Molecular weight, Number average and weight average molecular weight, Sedimentation and Viscosity average molecular weight, Molecular weight and degree of polymerization, Polydispersity, Size of polymer molecules.
4	**UNIT IV (08 Lecture)** Glass transition temperature, Transitions, significance, factors influencing the Tg. Effect of Plasticizers on Tg. Glass transition of copolymers. Morphology and order of Polymers, Crystallinity in polymers, Degree of crystallinity and Polymer crystallization. Effect of crystallinity on properties of Polymers.
5	**UNIT V (08 Lecture)** Polymer degradation and stability. Types of degradation. Mechanism of degradation. Factors affecting degradation. Polymer solutions. Process of dissolution of polymers. Thermodynamics of Dissolution. Flory-Huggins Theory, Viscosity of Polymeric solutions.

Books and References:

1. Plastics Material, Brydson, J.A
2. Text Book of Polymer Science, Billmeyer, Fred W.
3. Principles of Polymer Systems by Ferdinand Rodriguez
4. Principles of Polymer Chemistry by A. Ravve
5. Introduction of Polymer Science by Hans-Georg Elias
6. Polymer Science & Technology by Joel R. Fried.
7. Polymer Science by Gowariker V R, Vishwanathan NV, Jayadev Sreedhar.

BPL451- SYNTHESIS AND POLYMERIZATION LAB		
Course Outcome: Student will be able to		**Blooms Taxonomy**
CO 1	Synthesize a Polymer sample by various Polymerization Techniques	**K3**
CO 2	Analyze the estimation of Molecular weight of Polymers	**K3**
CO 3	Synthesize Thermoset Plastics by simple chemical methods	**K3**
CO 4	Depolymerize a waste plastic at laboratory level	**K2**
CO 5	Synthesize a co-polymer by simple chemical method	**K2**

Minimum eight experiments out of the following:

1. Synthesis of a polymer by Bulk polymerization Techniques
2. Synthesis of a polymer by Solution polymerization Techniques
3. Synthesis of a polymer by Suspension polymerization Techniques
4. Synthesis of a polymer by Emulsion polymerization Techniques
5. Determination of Molecular weight by viscosity.
6. Preparation of Phenol Formaldehyde resin
7. Preparation of Urea Formaldehyde resin
8. Preparation of Unsaturated Polyester resin
9. Preparation of saturated polyester resin
10. Depolymerization of waste thermoplastics such as polystyrene or poly-methyl methacrylate.
11. Synthesis of a copolymer based on any common monomers like styrene, acrylates, maleic anhydride, acrylic acid and methacrylic acid

BPL452- PLASTICS PROCESSING LAB		
Course Outcome: Student will be able to		**Blooms Taxonomy**
CO 1	Learn the working of Extrusion Machine and its various controls.	**K2**
CO 2	Learn the working of Injection Moulding machines and it process parameters	**K2**
CO 3	Learn the working of Compression & Transfer Moulding machines	**K2**
CO 4	Learn the working of Blow moulding, Thermoforming and Rotational Moulding processes	**K2**
CO 5	Analyze and understand how plastics are sealed and welded	**K3**
CO 6	Learn the working of Calendaring Process & Two Roll mill	**K2**

Minimum eight experiments out of the following:

1. Extrusion Process – Free sketch of Machine, Study of Parts & their function. Practice on Die setting, Cycle time analysis, Start up and shut down Procedure.
2. Twin screw extrusion process, Study of Parts & their function. Practice on Die setting, Cycle time analysis, Start up and shut down Procedure.
3. Injection Moulding Process - Free sketch of Machine, Study of Parts & their function. Cycle time analysis, Start up and shut down Procedure.
4. Compression Moulding or Transfer Moulding Process – Free sketch of Machine, Study of Parts & their function. Cycle time analysis, Start up and shut down Procedure.
5. Blow moulding Process – Free sketch of Machine, Study of Parts & their function, Parison die. Practice on Die centering, Cycle time analysis, Start up and shut down Procedure.
6. Thermoforming (Vacuum forming) Process – Free sketch of Machine, Study of Parts & their function. Cycle time analysis, Start up and shut down Procedure.
7. Rotational Moulding Process – Free sketch of Machine, Study of Parts & their function. Cycle time analysis, Start up and shut down Procedure.
8. Plastics Sealing Process – Free sketch of Machine, Study of Parts & their function. Cycle time analysis, Start up and shut down Procedure.
9. Plastics welding Process – Free sketch of Machine, Study of Parts & their function. Cycle time analysis, Start up and shut down Procedure.
10. Study of various parts of a Calendaring process. Start and shut down procedure.
11. Study of various parts of a Two –roll mill. Preparation of a sheet on Two Roll Mill.

BPL453- IDENTIFICATION AND ANALYSIS OF POLYMERS LAB		
Course Outcome: Student will be able to		**Blooms Taxonomy**
CO 1	Recognize various types of Plastics	**K3**
CO 2	Recognize various types of Rubbers	**K3**
CO 3	Analyze the estimation of Molecular weight of Polymers	**K3**
CO 4	Determine the Epoxy equivalent and acid value	**K2**
CO 5	Determine the K-Value of PVC resin	**K2**

Minimum eight experiments out of the following:

1. Identification of polymers by simple methods like density, melting point, burning characteristics, solubility, and confirmatory test by chemical analysis.
   Plastics: (Any three)
   a. Polyethylene
   b. Polypropylene
   c. Polystyrene
   d. Polyvinyl Chloride
   e. Polyamide
   Rubbers: (Any three)
   f. Natural Rubber (NR)
   g. Polybutylene Rubber (BR)
   h. Styrene Butadiene Rubber (SBR)
   i. Isoprene Rubber (IR)
   j. Ethylene Propylene Diene Monomer (EPDM)
2. Determination of molecular weight by end group analysis (COOH group)
3. Determination of molecular weight of polymers by viscosity method.
4. Determination of epoxy equivalent.
5. Determination of acid value of polyester resin.
6. Determination of K – value of PVC resin